

stpois: Fast Poisson event-history regression with high-dimensional fixed effects

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Abstract Poisson regression on episode-split survival data is a workhorse of event-history analysis, but two practical obstacles limit its use with full-population register data: models with rich fixed-effect structures become slow or infeasible to estimate with explicit dummy variables, and continuous covariates prevent the collapsing of individual records to risk cells that otherwise makes such models fast. This article introduces the community-contributed command `stpois`, which fits Poisson event-history models directly on `stset` data through three estimation paths: an exact path that wraps `poisson` with automatic exposure; an exact high-dimensional fixed-effects path implemented via iteratively reweighted alternating projections in Mata, with OIM, robust, and cluster-robust standard errors that are numerically identical to the dummy-variable model; and two fast paths that retain continuous covariates while moving the heavy computation to collapsed risk cells. The first folds first-stage continuous effects into a mutated exposure and re-estimates on cells (exact coefficients, standard errors conditional on the first stage); the second fits the exact individual-level maximum likelihood by Newton iterations on exponentially tilted within-cell moments, with one cheap microdata pass per iteration and all remaining algebra at the cell level, reproducing full-model coefficients and standard errors to numerical precision at a fraction of the cost. I describe the methods, document their exactness properties, and illustrate the command with examples.

1 Introduction

The equivalence between piecewise-constant hazard models and Poisson regression on episode-split data is one of the older and more useful results in event-history analysis (Holford 1980; Laird and Olivier 1981; Whitehead 1980). Splitting each subject’s time at risk into episodes, treating the event indicator in each episode as a Poisson count, and entering the log of exposure time as an offset yields maximum-likelihood estimates of a proportional-hazards model with a hazard that is constant within episodes. In Stata, this model can be fitted with `poisson` after manually generating the exposure variable, or — for the single-episode exponential case — with `streg, distribution(exponential)`. The approach is standard in epidemiology (Clayton and Hills 1993; Carstensen 2007) and is covered at length by Rabe-Hesketh and Skrondal (2022).

Two practical obstacles arise when this framework meets modern administrative data. Full-population registers of the kind available in the Scandinavian countries routinely produce analysis files with millions or tens of millions of episode records, particularly after splitting on calendar time, age, and duration. First, models with rich fixed-effect structures — birth cohorts, municipalities, detailed occupational codes, or their interactions — become slow or infeasible when the fixed effects are entered as explicit

dummy variables, since estimation time and memory grow with the square of the parameter count. The `ppmlhdfc` command (Correia et al. 2020) solves this problem for general Poisson models, building on the alternating-projections approach of Guimarães and Portugal (2010), but it is an external dependency with its own installation chain, which matters in the locked-down secure environments where register data must typically be analyzed. Second, and less widely appreciated, the Poisson likelihood for purely categorical covariates depends on the data only through the sums of events and exposure within covariate cells. Estimation on the collapsed cell data is then *exact* and often several orders of magnitude faster. A single continuous covariate breaks this property, and analysts face an unattractive choice between discretizing the covariate and giving up the speed advantage entirely.

The command described in this article, `stpois`, addresses both obstacles within a single interface for `stset` data. It offers three estimation paths:

1. A **standard path** that wraps `poisson`, using the event indicator `_d` as outcome and time at risk `_t - _t0` as exposure. Coefficients and standard errors are identical to `streg, distribution(exponential)`.
2. An **exact high-dimensional fixed-effects path**, `absorb()`, implementing iteratively reweighted least squares with alternating-projections demeaning in Mata — the same mathematical approach as `ppmlhdfc`, but self-contained with no dependencies. OIM, robust, and cluster-robust standard errors are numerically identical to those from the corresponding dummy-variable model.
3. Two **fast paths**, `fast(offset)` and `fast(moments)`, which retain continuous covariates while moving the heavy computation to the collapsed cells. `fast(offset)` folds first-stage estimates of the continuous effects into a “mutated” exposure before collapsing and re-estimating; its coefficients are exact, its standard errors conditional on the first stage. `fast(moments)` fits the **exact** individual-level maximum likelihood — coefficients, log likelihood, and OIM, robust, and cluster-robust standard errors all identical to `poisson` — by Newton iterations that accumulate exponentially tilted within-cell moments of the continuous covariates in a single microdata pass per iteration and do all remaining algebra at the cell level.

I am explicit throughout about what is exact and what is not: of the four estimation routes, only the `fast(offset)` standard errors involve an approximation. Section 2 sets out the modeling framework, section 3 describes the collapsed-cell methods, and section 4 describes the fixed-effects algorithm. Sections 5 and 6 document the syntax and illustrate the command, and section 7 reports validation results and timings.

2 Poisson event-history regression

Consider `stset` data in which record i contributes an event indicator $d_i \in \{0, 1\}$ (Stata’s `_d`) and a time at risk $t_i = _t - _t0$. Under a proportional-hazards model with hazard rate $\lambda_i = \exp(\mathbf{x}'_i \boldsymbol{\beta})$ constant within records, the log likelihood for right-censored data is, up to a constant,

$$\ell(\boldsymbol{\beta}) = \sum_i \{d_i \mathbf{x}'_i \boldsymbol{\beta} + d_i \log t_i - t_i \exp(\mathbf{x}'_i \boldsymbol{\beta})\},$$

which is proportional to the likelihood of a Poisson model for d_i with mean $\mu_i = t_i \exp(\mathbf{x}'_i \boldsymbol{\beta})$, that is, with $\log t_i$ as offset. Hence any software that fits Poisson regression with an offset fits this hazard model, and duration dependence or period effects are accommodated by episode splitting (`stsplit`) followed by indicator variables for the resulting time bands (Holford 1980; Laird and Olivier 1981). The Poisson “counts” here are 0/1 event indicators; nothing requires the outcome to be a genuine count, and the pseudo-maximum-likelihood results of Gourieroux et al. (1984) imply that the coefficient estimates are consistent whenever the log-linear mean is correctly specified, with robust standard errors available in the usual way (see also Santos Silva and Tenreiro 2006).

`stpois` fits exactly this model on the current `stset` data. On the standard path it constructs the exposure variable, calls `poisson`, and posts results with survival-flavored headers (numbers of subjects and failures, time at risk). The point of the command, however, is not the convenience wrapper but the two departures from it, to which I now turn.

2.1 Exact collapsing with categorical covariates

When all covariates are categorical, the covariate vector takes a finite number of distinct values, indexing cells $j = 1, \dots, J$. The log likelihood above then depends on the data only through the within-cell sums $D_j = \sum_{i \in j} d_i$ and $T_j = \sum_{i \in j} t_i$:

$$\ell(\beta) = \sum_j \{D_j \mathbf{x}'_j \beta - T_j \exp(\mathbf{x}'_j \beta)\} + \text{const} .$$

(D_j, T_j) are sufficient statistics, and Poisson estimation on the J collapsed cells reproduces the individual-level maximum-likelihood estimates exactly — coefficients, standard errors, and all. With millions of episode records but a few hundred cells (period $\times$ education $\times$ region, say), the gain is dramatic, and this is the standard trick behind register-based rate modeling (Clayton and Hills 1993). A continuous covariate destroys the sufficiency because $\exp(\gamma x_i)$ no longer aggregates: $\sum_{i \in j} t_i \exp(\gamma x_i) \neq T_j \exp(\gamma \bar{x}_j)$ by Jensen's inequality. The two fast paths in `stpois` are two different responses to this problem.

3 Fast estimation with continuous covariates

Partition the covariates into continuous variables $\mathbf{X}_i$ with coefficients γ and categorical variables defining cells j with cell effects α_j , so that $\lambda_i = \exp(\gamma' \mathbf{X}_i + \alpha_{j(i)})$. In the `stpois` syntax, bare variable names are treated as continuous and `i.`-prefixed terms as categorical; the categorical variables define the collapse cells.

3.1 Two-stage offset: fast(offset)

The multiplicative structure of the log link, $\exp(\gamma' \mathbf{X}_i + \alpha_j) = \exp(\gamma' \mathbf{X}_i) \exp(\alpha_j)$, means that a *known* γ can be folded into the exposure. Stage 1 estimates $\hat{\gamma}$ by Poisson regression on the full individual-level data, including the categorical indicators to avoid omitted-variable bias. Stage 2 constructs the mutated exposure

$$\tilde{t}_i = t_i \exp(\hat{\gamma}' \mathbf{X}_i),$$

collapses d_i and $\tilde{t}_i$ to cell sums $(D_j, \tilde{T}_j)$, and fits the categorical-only Poisson model on the J cells with $\log \tilde{T}_j$ as offset.

Given $\hat{\gamma}$, this second stage is exact in the same sense as section 2.1: because the stage-1 model is the very model of interest, the stage-2 score equations for the cell parameters are the profile score equations of the full model, and the stage-2 coefficient estimates replicate the full maximum-likelihood estimates of the categorical parameters. In the validation exercise of section 7 they agree to six decimal places. What is *not* exact are the standard errors: the stage-2 variance matrix conditions on $\hat{\gamma}$ and therefore omits the first-stage sampling variation that a joint or two-step corrected estimator (Murphy and Topel 1985) would propagate. The constant's standard error is visibly too small in the examples below; the cell-contrast standard errors are much less affected because the first-stage error largely shifts all cells together. Since stage 1 runs on the full individual-level data, `fast(offset)` does not save time in a single estimation — its value lies in situations where the collapsed data are reused, as when many categorical specifications are explored over a fixed set of continuous controls, or when the collapsed file, unlike the microdata, may leave a secure server.

3.2 Iterated cell moments: fast(moments)

The second method dispenses with the two-stage logic and fits the exact model directly, exploiting the cell structure to make each Newton iteration cheap. Write the full parameter vector as $\theta = (\gamma, \delta)$, where δ collects the coefficients on the cell-level design $\mathbf{w}_j$ (the categorical dummies and the constant), so that $\mu_i = t_i \exp(\gamma' \mathbf{X}_i + \delta' \mathbf{w}_{j(i)})$. The score and observed information of the Poisson likelihood then decompose into pieces that are either fixed cell sums or *exponentially tilted within-cell moments* of the continuous covariates:

$$M_j = \sum_{i \in j} \mu_i, \quad \mathbf{m}_j = \sum_{i \in j} \mu_i \mathbf{X}_i, \quad \mathbf{Q} = \sum_i \mu_i \mathbf{X}_i \mathbf{X}_i'.$$

The score is $(\sum_i d_i \mathbf{X}_i - \sum_j \mathbf{m}_j, \sum_j \mathbf{w}_j (D_j - M_j))$ and the information matrix has blocks $\mathbf{Q}$, $\sum_j \mathbf{m}_j \mathbf{w}_j'$, and $\sum_j M_j \mathbf{w}_j \mathbf{w}_j'$. Each Newton iteration therefore requires one pass over the microdata to accumulate the tilted moments — at cost $O(np^2)$, with p the number of *continuous* covariates only — after which everything happens on the J cells. A full-model Newton step in `poisson`, by contrast, costs $O(n(p+k)^2)$ with k the number of categorical parameters, so the per-iteration saving is a factor of roughly $\{(p+k)/p\}^2$: about a hundredfold for two continuous covariates and eighteen categorical terms. Step-halving on the log likelihood guards the updates, and convergence is declared when $\max |\Delta \theta| / (1 + |\theta|)$ falls below `tolerance()`.

Because this is Newton on the exact likelihood, nothing is approximated: coefficients, the log likelihood, and the inverse-information (OIM) standard errors are numerically identical to the full `poisson` model, and robust and cluster-robust variance estimators are computed from the same per-observation scores as their `poisson` counterparts (HC1 and $G/(G-1)$ corrections included). The test suite verifies agreement to 10^{-6} on all of these. In particular — and in contrast to a pure collapsed-data method — the continuous coefficients γ are identified from the full individual-level variation, not merely from between-cell differences in cell means; `fast(moments)` estimates them with full efficiency even when the continuous covariates are orthogonal to the cell structure. An earlier version of this method entered second-order cumulant-generating-function corrections as free regressors in a collapsed Poisson model; the Newton formulation supersedes it, and the `skewness` option that supported higher-order corrections is retained only for backward compatibility (it is ignored with a note).

Since exact estimation is available at collapsed-cell speed, `fast(moments)` is the recommended fast path. `fast(offset)` remains useful for its by-product: the mutated-exposure cell file is itself a valid dataset for categorical-only estimation, which matters when only aggregated data may leave a secure environment.

4 High-dimensional fixed effects: absorb()

The `absorb(varlist)` option fits the exact model with one or more sets of fixed effects without creating dummy variables, using iteratively reweighted least squares (IRLS) combined with weighted alternating-projections demeaning — the approach introduced to Stata by Guimarães and Portugal (2010) and refined by Correia et al. (2020). Each IRLS iteration:

1. forms the working variable $z_i = \eta_i + (d_i - \mu_i)/\mu_i$ and weights $w_i = \mu_i$, where η_i is the current linear predictor including the offset and $\mu_i = t_i \exp(\eta_i)$;
2. demeans z and each covariate within the groups of each absorbed variable in turn, iterating this Gauss–Seidel cycle to convergence (weighted alternating projections);
3. regresses the demeaned $\tilde{z}$ on the demeaned $\tilde{\mathbf{X}}$ by weighted least squares to update β ; and
4. recovers the fixed-effect contribution as the complement of the projection and updates η .

Convergence is declared when $\max_k |\Delta \beta_k| / (1 + |\beta_k|)$ falls below `tolerance()` (default 10^{-8}). The implementation is pure Mata with no dependencies.

Standard errors come from the partitioned information at convergence: $V = (\tilde{\mathbf{X}}'W\tilde{\mathbf{X}})^{-1}$ for `vce(oim)`, the HC1 sandwich with meat $\sum_i s_i s_i' \cdot n/(n-1)$ and scores $s_i = (d_i - \mu_i)\tilde{\mathbf{x}}_i$ for `vce(robust)`, and the $G/(G-1)$ -corrected clustered sandwich with scores summed within clusters for `vce(cluster clustvar)`. By the Frisch–Waugh–Lovell logic these are not approximations: all three match the corresponding dummy-variable `poisson` output at numerical precision, which the test suite verifies (section 7).

Groups of an absorbed variable in which no events occur are separated in the maximum-likelihood sense — their fixed effects diverge to $-\infty$ — and `stpois` drops such observations before estimation with a message, mirroring the practice of `ppmlhdfc` (Correia et al. 2020, who treat separation in far greater generality). The inner demeaning loop costs $O(n)$ per absorbed variable and pass, which is efficient for the moderate group counts typical of event-history applications (periods, cohorts, regions, occupations). For extreme cases — say, person fixed effects with millions of levels — `ppmlhdfc`’s accelerated solver remains the better tool, and `stpois` makes no attempt to compete there.

5 The `stpois` command

5.1 Syntax

```
stpois varlist [if] [in] [weight] [, irr nolog level(#) noconstant
      vce(vcetype) robust cluster(varname)
      absorb(varlist) tolerance(#) maxiter(#)
      fast(offset | moments) skewness poisson_options ]
```

The data must be `stset`; see [ST] `stset`. `fweights`, `iweights`, and `pweights` are allowed. `fast()` and `absorb()` may not be combined. In the fast paths, bare variable names are treated as continuous and `i.`-prefixed variables as categorical grouping variables; interactions are not supported there.

5.2 Options

irr reports exponentiated coefficients (incidence-rate ratios, equal to hazard ratios in this model). Display only.

nolog suppresses iteration logs, including the progress notes of the fast and HDFE paths.

level(#) sets the confidence level; the default is `level(95)`.

noconstant suppresses the constant term.

vce(vcetype), **robust**, and **cluster(varname)** specify the variance estimator. On the standard path these are passed through to `poisson` and accept anything `poisson` accepts. With `absorb()` and `fast(moments)`, `vce(oim)` (the default), `vce(robust)`, and `vce(cluster clustvar)` are supported, all numerically exact relative to the corresponding full `poisson` model.

absorb(varlist) absorbs the fixed effects of one or more categorical variables via IRLS with alternating-projections demeaning; see section 4.

tolerance(#) and **maxiter(#)** control the iteration convergence criterion and cap for `absorb()` and `fast(moments)`; defaults are `1e-8` and `100`.

fast(offset | moments) selects a collapsed-cell fast path; see section 3. **skewness** is obsolete and ignored with a note (it supported higher-order corrections in a superseded approximate version of `fast(moments)`).

Any remaining options on the standard path are passed to `poisson`.

5.3 Stored results

`stpois` stores the usual `e(b)`, `e(V)`, `e(sample)`, log likelihood, and model-test scalars, plus survival-specific scalars `e(N_sub)` (subjects), `e(N_fail)` (failures), and `e(risk)` (total time at risk). The fast paths add `e(N_cells)` and the macros `e(cont_vars)` and `e(cat_vars)`; `absorb()` adds `e(absorb)` and, with clustering, `e(N_clust)` and `e(clustvar)`. `e(predict)` is set to `stpois_p`, so `predict` offers the linear predictor (`xb`, the default, including the log-exposure offset unless `nooffset`), predicted numbers of events (`n`), hazard rates (`hazard` or `hr`), and survival probabilities (`survival`). Standard postestimation commands (`margins`, `lincom`, `test`, `estimates`) work as usual. The standard path, `absorb()`, and `fast(moments)` all report the individual-level log likelihood; `fast(offset)` reports the collapsed-cell likelihood of its second stage, which differs by a data-dependent constant and should not be compared across paths.

6 Examples

6.1 Standard path

The default path reproduces `streg`, `distribution(exponential)` exactly (the log likelihoods differ by a constant because `streg` includes the $d_i \log t_i$ terms):

```
. webuse stan3
. stset t1, failure(died) id(id)
. stpois age posttran, nolog
```

Poisson event-history regression

No. of subjects =	103	Number of obs =	172
No. of failures =	75		
Time at risk =	31938	LR chi2(2) =	36.75
Log likelihood =	-233.83404	Prob > chi2 =	0.0000

	_d	Coefficient	Std. err.	z	P> z	[95% conf. interval]
	age	.0605817	.015235	3.98	0.000	.0307216 .0904417
	posttran	-1.285206	.2377048	-5.41	0.000	-1.751099 -.8193135
	_cons	-7.8376	.7094787	-11.05	0.000	-9.228153 -6.447047

6.2 Absorbing fixed effects

`absorb()` returns the same coefficients and standard errors as the explicit dummy-variable model, without estimating the absorbed effects:

```
. stpois age posttran, absorb(surgery) nolog
```

Poisson EHA – HDFE (absorb: surgery)

No. of subjects =	103	Number of obs =	172
No. of failures =	75		
Time at risk =	31938	LR chi2(2) =	49.18
Log likelihood =	-227.61949	Prob > chi2 =	0.0000

	<code>_d</code>	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
	+						
	age	.0610952	.014454	4.23	0.000	.0327659	.0894244
	posttran	-1.141144	.2402034	-4.75	0.000	-1.611934	-.6703538

For comparison, `poisson _d age posttran i.surgery, exposure(exp)` on the same sample gives `age` = .0610952 (SE .014454) and `posttran` = -1.141144 (SE .2402034): identical to all printed digits.

6.3 Fast paths on a large synthetic dataset

The following simulation generates 500,000 subjects in 200 risk cells (4 education × 5 region × 10 period), with two continuous covariates (`age`, `income`) that affect the hazard and are assigned independently of the cells — a deliberately demanding design for any method that leans on cell-level information:

```
. clear
. set obs 500000
. set seed 20260713
. gen edu = ceil(4 * runiform())
. gen region = ceil(5 * runiform())
. gen period = ceil(10 * runiform())
. gen age = 30 + 20 * runiform()
. gen income = rnormal(10, 2)
. gen lambda = exp(-6 + 0.03*age + 0.05*income
> + 0.3*(edu==2) + 0.5*(edu==3) - 0.2*(edu==4)
> + 0.1*(region==2) - 0.2*(region==3) + 0.2*(period >= 6))
. gen ptime = 0.5 + 0.5 * runiform()
. gen failed = (runiform() < 1 - exp(-lambda * ptime))
. gen id2 = _n
. stset ptime, failure(failed) id(id2)
```

This yields 6,759 events. Estimating the full model and both fast paths:

```
. stpois age income i.edu i.region i.period, nolog
. stpois age income i.edu i.region i.period, fast(offset) nolog
. stpois age income i.edu i.region i.period, fast(moments) nolog
```

The table below extracts selected coefficients (standard errors in parentheses):

	Full MLE	fast(offset)	fast(moments)
age (γ_1)	0.0265 (0.0021)	— (stage 1)	0.0265 (0.0021)
income (γ_2)	0.0518 (0.0061)	— (stage 1)	0.0518 (0.0061)
2.edu	0.3071 (0.0353)	0.3071 (0.0353)	0.3071 (0.0353)
3.edu	0.5094 (0.0339)	0.5094 (0.0339)	0.5094 (0.0339)
3.region	-0.1654 (0.0401)	-0.1654 (0.0401)	-0.1654 (0.0401)
9.period	0.2175 (0.0538)	0.2175 (0.0538)	0.2175 (0.0538)

	Full MLE	fast(offset)	fast(moments)
Log likelihood	-35297.233	(cell scale)	-35297.233
Estimation time	1.35 s	1.42 s	0.62 s

Three features of this table repay attention. First, `fast(moments)` reproduces the full maximum-likelihood output — every coefficient, every standard error, and the log likelihood — to all printed digits, in five Newton iterations, at less than half the estimation time; the advantage grows with the number of categorical parameters and with the size of the microdata. This holds even though the continuous covariates were assigned independently of the cells: identification comes from the individual-level likelihood, not from between-cell variation. Second, `fast(offset)` also reproduces the categorical coefficients to six decimal places, as the sufficiency argument of section 3.1 predicts, and its cell-contrast standard errors are correct to the printed digits; only the constant’s standard error (not shown) is understated, since it conditions on the stage-1 estimates. Third, `fast(offset)` saves no time in a single estimation — its stage 1 runs on the microdata — which is why `fast(moments)` is the recommended fast path unless the collapsed cell file itself is needed.

6.4 Episode-split register-style workflow

In applications, the typical sequence splits spells on calendar time or age and then estimates piecewise-constant hazard models:

```
. stsplit period, at(1990(5)2020) after(time=entry)
. stpois i.period age income i.edu, irr
. stpois age income, absorb(period edu region) vce(cluster region)
. stpois age income i.period i.edu i.region, fast(moments)
```

7 Validation and performance

The package ships with a test suite (`tests/test_stpois.do`) that runs on the `stan3` heart-transplant data and on simulated data. It verifies that (i) the standard path converges and matches `streg, distribution(exponential)`; (ii) `absorb()` matches the explicit dummy-variable `poisson` model on coefficients *and* standard errors to within 10^{-4} (the observed agreement is on the order of 10^{-8}), for OIM and robust variance estimators, with the cluster estimator covered by the same sandwich code path; (iii) `fast(offset)` reproduces full-MLE categorical coefficients on collapsed cells; and (iv) `fast(moments)` matches the full `poisson` model on coefficients, standard errors (OIM, robust, and clustered), and log likelihood to within 10^{-6} , including in a two-cell setting where the continuous coefficient is identified purely from within-cell variation. All tests pass on Stata 18/MP; the command requires version 14.

The timings in section 6.3 were obtained on a single run (Stata MP, Apple silicon) and should be read as indicative. The general pattern is robust, however: the `fast(moments)` cost per Newton iteration is $O(np^2)$ in the number of *continuous* covariates only, versus $O(n(p+k)^2)$ for `poisson` with k categorical parameters, so the advantage grows quadratically in the richness of the categorical structure. For register applications with 10^7 or more episodes and detailed period $\times$ education $\times$ region interactions, this turns minutes into seconds, which changes what is practical to do interactively: specification searches over the categorical structure, bootstrap loops, or estimation inside secure environments with tight session limits.

8 Conclusion

`stpois` packages three things that belong together but have hitherto required three different workflows: the Poisson formulation of event-history models on `stset` data, exact absorption of high-dimensional

fixed effects without external dependencies, and cell-accelerated estimation that retains continuous covariates. Three of the four estimation routes are exact in the strict sense — the test suite holds the `absorb()` and `fast(moments)` variance estimators to the full-model benchmark, not merely the coefficients — and the one remaining approximation, the `fast(offset)` standard errors, is documented rather than hidden; a Murphy–Topel correction for it fits the existing architecture and is the natural extension. The command, its help file, and the test suite are available from the author’s repository and are installable with `net install`.

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